

What is the problem?

Airplane crashes are things that people tend to be very afraid of, even though flying is generally much safer than people think. What interests me about this topic is that when an accident does happen, the outcome can be very different depending on the situation. Some accidents have no fatalities, while others result in many people dying. An aviation company or safety organization could use historical crash data to better understand what conditions are connected to more serious outcomes.

For this project, the business problem is to determine whether an airplane accident is likely to result in at least one fatality based on information known about the accident.

Why is this problem important?

This could help aviation organizations identify patterns connected to more dangerous accidents. If certain aircraft types, operators, locations, time periods, or flight conditions appear more often in fatal accidents, safety teams could pay closer attention to those areas when reviewing risks and planning safety improvements.

Who is affected by this problem?

Airline and aviation companies, pilots, maintenance teams, safety investigators, and other aviation organizations could all be affected. Passengers are obviously the most important group because their safety is directly involved, but companies also have to think about training, aircraft safety, procedures, and preventing future accidents.

What data is needed?

I would use the historical airplane crash dataset provided for this assignment. It contains 5,268 records and 13 columns, including Date, Time, Location, Operator, Flight #, Route, Type, Registration, Aboard, Fatalities, Ground, and Summary.

The most useful information would be the location, operator, route, aircraft type, number of people aboard, and other details that could describe the accident.

Where would the data come from?

The main source would be the historical airplane crash dataset provided for this project. If this were being used by an actual aviation company, it could also be combined with company safety records, aircraft maintenance records, weather information, flight information, and other aviation databases.

How much historical data is needed?

I would use as much of the historical dataset as possible because it contains thousands of accident records covering a long period of aviation history. Having many years of information would give the model more examples to learn from instead of only looking at a small number of recent accidents.

What variables are important?

Some important variables would be:

- Aboard: number of people on the aircraft
- Type: aircraft type
- Operator: company or organization operating the aircraft
- Location: where the accident happened
- Route: where the aircraft was traveling
- Ground: number of people on the ground who were killed

Is there missing or inaccurate data?

Yes. Some columns have a significant amount of missing information. For example, the Time, Flight #, Route, Registration, and Summary columns are not complete for every accident. This matters because a model could have trouble learning from a variable when many records do not contain that information.

Features

The features I would use are:

1. Aboard: how many people were on the aircraft
2. Aircraft Type: the type/model of aircraft involved
3. Operator: the company or organization operating the aircraft
4. Location: where the accident occurred

5. Date or Year: when the accident occurred
6. Time of Day: the time of the accident when available
7. Route: the planned route when available

These features give information about the aircraft, operation, and circumstances surrounding the accident.

Target

The target would be:

Fatal accident: Yes or No

I would define this using the Fatalities column. If Fatalities is greater than 0, the target would be Yes. If Fatalities is 0, the target would be No.

I would not use the Fatalities column itself as a feature because it is what the model is trying to predict. Using it as an input would basically give the model the answer ahead of time.

Part 4: Select an AI/ML Approach

Supervised or Unsupervised Learning?

I would use supervised learning because the historical data already contains known outcomes. The dataset tells us how many fatalities occurred in each accident, so we can use that information to create the target of fatal or nonfatal. The model would learn from past examples where the outcome is already known. After learning those patterns, it could use the other information about an accident to classify a new accident. This makes supervised learning a better choice than unsupervised learning for this problem.

Classification, Prediction, Both, or Neither?

I would use classification because the result has two categories: fatal accident or nonfatal accident. The model would not need to give a specific number of fatalities as its main output. Instead, it would place an accident into one of the two categories based on the features. This is useful because the business question is focused on whether an accident results in fatalities, rather than predicting the exact number of people who would die.

Data Quality Problem 1: Missing values

Some important fields are missing from the dataset. For example, many records do not have a recorded time, flight number, or route.

Business impact: If important information is missing, the model may have less information to work with and its results could be less reliable.

Data Quality Problem 2: Inconsistent information

The dataset covers more than a century of aviation accidents, so information may have been recorded differently depending on the time period. Aircraft names, operator names, locations, and other fields may not always follow the same format.

Business impact: The model could treat information that should be grouped together as different values. That could make patterns harder to identify.

Data Quality Problem 3: Very different amounts of information across records

Some accident records have detailed summaries while others have much less information. There are also fields with many missing entries.

Business impact: Accidents with incomplete information may be harder for the model to classify correctly. An aviation company would need to be careful about trusting a prediction when the information going into the model is incomplete.



Business Problem

An aviation organization wants to use historical airplane crash data to identify whether an accident is likely to involve fatalities and understand which factors are associated with more serious accidents.

Data Needed

The model would use historical accident information such as aircraft type, operator, location, date, time, route, and number of people aboard. The dataset provided contains 5,268 accident records.

AI/ML Approach

I would use supervised machine learning because the historical data includes known accident outcomes.

Type of Learning

The problem is a classification problem because the model would classify an accident as either fatal or nonfatal.

Model Output

The model would output:

Fatal accident: Yes or No

Business Benefit

The results could help aviation organizations understand patterns related to serious accidents and decide where more safety attention may be needed. I think this would be useful because it turns a large amount of historical accident information into something that could be easier for a safety team to interpret and use when reviewing risks.

Potential Risk

The biggest risk is that historical data is not always complete or consistent. A model could also find patterns in old aviation accidents that do not apply as well to modern aircraft and safety systems. Because of that, the model should be used as a tool for supporting safety decisions, not as the only source of information.

Generative AI Use: I used generative AI, specifically Claude by Anthropic, after completing the assignment to refine my writing and correct grammar, punctuation, and capitalization mistakes. I type quickly, so I used it to clean up those errors without changing my ideas or the content of my work.